

由于程序实现中正则化引起的迭代方向不够精确的问题，考虑将核心方程组化为 2x2 分块已经得到的计算公式：

$$z_j = z_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} - (s_j - \rho\xi_j) \right)$$

$$y_j = y_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} + (s_j - \rho\xi_j) \right)$$

$$\psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k, \xi_k) = \begin{bmatrix} r_k^d \\ r_k^h \\ r_k^e \\ r_k^\xi \end{bmatrix} = \begin{bmatrix} Px_k + c + A^\top t_k + G^\top s_k \\ Ax_k - b \\ z_k - \xi_k \\ Gx_k + \xi_k - h \end{bmatrix} \quad \text{size:} \quad \begin{bmatrix} n \\ p \\ m \\ m \end{bmatrix}$$

$$\phi_{(\mu, \rho)}(x, t, s, \xi) = \frac{1}{2} \|\psi_{(\mu, \rho)}(x, t, s, \xi)\|^2$$

$$\Delta\mu = -\mu_k + \gamma\phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k, \xi_k)$$

$$\begin{bmatrix} P & A^\top & G^\top \\ A & O & O \\ G & O & -\frac{1}{\rho_k} Y_k^{-1} Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = - \begin{bmatrix} r_k^d \\ r_k^h \\ (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e + r_k^\xi \end{bmatrix}$$

$$d_\xi = -r_k^\xi - Gd_x$$

处理 3x3 分块矩阵：

第三行乘以 $\rho_k G^\top Z_k^{-1} Y_k$ 加到第一行：

$$\begin{bmatrix} P + \rho_k G^\top Z_k^{-1} Y_k G & A^\top & O \\ A & O & O \\ G & O & -\frac{1}{\rho_k} Y_k^{-1} Z_k \end{bmatrix} \begin{bmatrix} d_x \\ d_t \\ d_s \end{bmatrix} = - \begin{bmatrix} r_k^d + \rho_k G^\top Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu G^\top Z_k^{-1} e + \rho_k G^\top Z_k^{-1} Y_k r_k^\xi \\ r_k^h \\ (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e + r_k^\xi \end{bmatrix}$$

提取出第三行：

$$\begin{bmatrix} P + \rho_k G^\top Z_k^{-1} Y_k G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} = - \begin{bmatrix} r_k^d + \rho_k G^\top Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu G^\top Z_k^{-1} e + \rho_k G^\top Z_k^{-1} Y_k r_k^\xi \\ r_k^h \end{bmatrix}$$

$$Gd_x - \frac{1}{\rho_k} Y_k^{-1} Z_k d_s + (Z_k + Y_k) Y_k^{-1} r_k^e + \frac{1}{\rho_k} \Delta\mu Y_k^{-1} e + r_k^\xi = 0$$

$$\rho_k Gd_x + \rho_k (Z_k + Y_k) Y_k^{-1} r_k^e + \Delta\mu Y_k^{-1} e + \rho_k r_k^\xi = Y_k^{-1} Z_k d_s$$

$$d_s = \rho_k Y_k Z_k^{-1} Gd_x + \rho_k Z_k^{-1} (Z_k + Y_k) r_k^e + \Delta\mu Z_k^{-1} e + \rho_k Y_k Z_k^{-1} r_k^\xi$$

$$\text{代入 } Y_k^{-1} = \frac{\rho_k}{\mu_k} Z_k, \quad Z_k^{-1} = \frac{\rho_k}{\mu_k} Y_k$$

$$\begin{aligned}
& \begin{bmatrix} P + \frac{\rho_k^2}{\mu_k} G^\top Y_k^2 G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} \\
& = - \begin{bmatrix} r_k^d + \rho_k G^\top r_k^e + \frac{\rho_k^2}{\mu_k} G^\top Y_k^2 r_k^e + \frac{\Delta\mu\rho_k}{\mu_k} G^\top Y_k e + \frac{\rho_k^2}{\mu_k} G^\top Y_k^2 r_k^\xi \\ r_k^h \end{bmatrix} \\
& d_s = \frac{\rho_k^2}{\mu_k} Y_k^2 G d_x + \rho_k r_k^e + \frac{\rho_k^2}{\mu_k} Y_k^2 r_k^e + \frac{\Delta\mu\rho_k}{\mu_k} Y_k e + \frac{\rho_k^2}{\mu_k} Y_k^2 r_k^\xi
\end{aligned}$$

$$\begin{aligned}
& \begin{bmatrix} P + \frac{\rho_k^2}{\mu_k} G^\top Y_k^2 G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} = - \begin{bmatrix} r_k^d + G^\top \left(\rho_k r_k^e + \frac{\rho_k^2}{\mu_k} Y_k^2 (r_k^e + r_k^\xi) + \frac{\Delta\mu\rho_k}{\mu_k} Y_k e \right) \\ r_k^h \end{bmatrix} \\
& d_s = \frac{\rho_k^2}{\mu_k} Y_k^2 (G d_x + r_k^e + r_k^\xi) + \rho_k r_k^e + \frac{\Delta\mu\rho_k}{\mu_k} Y_k e
\end{aligned}$$

由于左上角的分块较为复杂，不好确定稀疏结构，先用实心的结构处理。

汇总公式：

$$z_j = z_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} - (s_j - \rho\xi_j) \right)$$

$$y_j = y_j(\xi_j, s_j; \mu, \rho) = \frac{1}{2\rho} \left(\sqrt{(s_j - \rho\xi_j)^2 + 4\rho\mu} + (s_j - \rho\xi_j) \right)$$

$$\psi_{(\mu_k, \rho_k)}(x_k, t_k, s_k, \xi_k) = \begin{bmatrix} r_k^d \\ r_k^h \\ r_k^e \\ r_k^\xi \end{bmatrix} = \begin{bmatrix} P x_k + c + A^\top t_k + G^\top s_k \\ A x_k - b \\ z_k - \xi_k \\ G x_k + \xi_k - h \end{bmatrix} \quad \text{size: } \begin{bmatrix} n \\ p \\ m \\ m \end{bmatrix}$$

$$\phi_{(\mu, \rho)}(x, t, s, \xi) = \frac{1}{2} \|\psi_{(\mu, \rho)}(x, t, s, \xi)\|^2$$

$$\Delta\mu = -\mu_k + \gamma\phi_{(\mu_k, \rho_k)}(x_k, t_k, s_k, \xi_k)$$

$$\begin{aligned}
& \begin{bmatrix} P + \frac{1}{\mu_k} G^\top (\rho_k Y_k)^2 G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} d_x \\ d_t \end{bmatrix} \\
& = - \begin{bmatrix} r_k^d + G^\top \left(\frac{1}{\mu_k} (\rho_k Y_k)^2 (r_k^e + r_k^\xi) + \rho_k r_k^e + \frac{\Delta\mu}{\mu_k} \rho_k Y_k e \right) \\ r_k^h \end{bmatrix}
\end{aligned}$$

$$d_s = \frac{1}{\mu_k} (\rho_k Y_k)^2 G d_x + \frac{1}{\mu_k} (\rho_k Y_k)^2 (r_k^e + r_k^\xi) + \rho_k r_k^e + \frac{\Delta\mu}{\mu_k} \rho_k Y_k e$$

$$d_\xi = -r_k^\xi - G d_x$$

$$W = G^\top (\rho_k Y_k)^2 G = (\rho_k Y_k G)^\top (\rho_k Y_k G)$$

$$W_{ij} = \sum_t (\rho_k Y_k G)_{ti} (\rho_k Y_k G)_{tj} = \sum_t \rho_k^2 y_{kt}^2 G_{ti} G_{tj}$$

$\rho_k Y_k$ 成对出现，可以考虑化简计算； $r_k^e + r_k^\xi = Gx_k + z_k - h$

需要考虑的问题：将 $\frac{1}{\mu_k} G^\top (\rho_k Y_k)^2 G$ 加到 P 上时只保留上三角的部分！ 需要把行列讨论清楚。

初始化部分需要适应新的 K 矩阵：

$$\begin{bmatrix} P & A^\top & G^\top \\ A & O & O \\ G & O & -I \end{bmatrix} \begin{bmatrix} x \\ t \\ s \end{bmatrix} = \begin{bmatrix} -c \\ b \\ h \end{bmatrix}$$

$$\begin{bmatrix} P + G^\top G & A^\top & O \\ A & O & O \\ G & O & -I \end{bmatrix} \begin{bmatrix} x \\ t \\ s \end{bmatrix} = \begin{bmatrix} -c + G^\top h \\ b \\ h \end{bmatrix}$$

推导出：

$$\begin{bmatrix} P + G^\top G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} = \begin{bmatrix} -c + G^\top h \\ b \end{bmatrix}$$

$$s = Gx - h$$

$$\xi = h - Gx$$

这里要令 $\mu_k = 1, \rho_k Y_k = I$ (便于 iterative refinement)，在计算完初始点之后重新设 μ_k, ρ_k 的值。

iterative refinement 要计算 $r - Kd$ (K 未正则化)

$$\begin{bmatrix} P + \frac{1}{\mu_k} G^\top (\rho_k Y_k)^2 G & A^\top \\ A & O \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} = \begin{bmatrix} \left(P + \frac{1}{\mu_k} G^\top (\rho_k Y_k)^2 G \right) x + A^\top t \\ Ax \end{bmatrix}$$

需要注意 data $\rightarrow P$ 是已经正则化的，要减去相应的量